

The complete ES defect atlas as affine torsion monodromy Source reconstruction, primitive cohomology, and all odd-level genera

The Clankers research workbench

Assistant-developed continuation; source-specific attribution retained

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Abstract

The full finite Erdős–Straus divisor atlas has a canonical integer return denominator at every labelled candidate. We identify its cyclic obstruction group with a primitive cohomology class of exact order D in the actual rank-four period monodromy circulated with the proposed S^6 construction. The affine slice has three residue coordinates. We classify all its orbits and all compactified genera for every odd level, retaining prime-power budgets, raw orders and candidate labels. A source-derived central commutator is translation by -12 . A counterexample prime would give a full sectionless decorated covering; no such prime is found. We distinguish this construction from the source’s six-dimensional total space, exhibit an obstruction to one proposed identification, and keep all source-global claims outside the new proofs.

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1 The source representation and its affine slice

We use the following integral matrices on the ordered rank-four lattice $\mathbb{L} = \mathbb{Z}\hat{\gamma} \oplus \mathbb{Z}\hat{u} \oplus \mathbb{Z}\hat{w} \oplus \mathbb{Z}\hat{\delta}$:

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & -1 & -1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -6 & 1 & 0 & 0 \\ 3 & 0 & 1 & 1 \end{pmatrix}, \quad A_\infty = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}. \quad (1)$$

They are the inverse-transpose monodromies in the circulated $(3, 4, \infty)$ period construction. The source locator is [Source(2026), §2], Lemmas 2.4–2.7. Direct multiplication proves $A_1^3 = A_2^4 = A_1 A_2 A_\infty = I$. We distinguish $\mathbb{L}$ from the rank-24 Leech lattice. No global assertion that the source threefold is S^6 is used below.

The invariant covector γ reads the first coordinate. For a positive odd integer D , let

$$\mathcal{F}_D = \{(1, x, y, z) \in (\mathbb{Z}/D)^4\} \cong (\mathbb{Z}/D)^3.$$

The induced permutations, composed right to left, are

$$\begin{aligned} a_1(x, y, z) &= (6 + y, -6 - x - y, z - 2 + x), \\ a_2(x, y, z) &= (-y, x - 6, z + 3 + y), \\ a_\infty(x, y, z) &= (x, y + x, z - 1), \end{aligned} \quad (\text{mod } D). \quad (2)$$

Their inverses are a_1^2, a_2^3 , and $(x, y, z) \mapsto (x, y - x, z + 1)$. The unique residue class modulo one is allowed; $\mathcal{F}_1$ is a singleton.

The base marking used earlier in the Fable comparison is reached by the explicit projective change $s = 1/2 - t$, with inverse $t = 1/2 - s$. It sends $(0, 1, \infty)$ to $(1/2, -1/2, \infty)$ and transports the three loop actions with those labels. It changes neither the modulus nor the monodromy representation. It is not a substitution into a Mellin integral.

Theorem 1.1 (Primitive obstruction class). *Write $A_g = \begin{pmatrix} 1 & 0 \\ b_g & B_g \end{pmatrix}$ for $g \in \Delta = \langle g_1, g_2 \mid g_1^3 = g_2^4 = 1 \rangle$. Then $b_{gh} = b_g + B_g b_h$. The integral cocycle class $[b]$ has infinite order and spans a split $\mathbb{Z}$ -summand of $H^1(\Delta, \mathbb{Z}_B^3)$. Its reduction $[b_D] \in H^1(\Delta, (\mathbb{Z}/D)_B^3)$ has exact additive order D . It restricts to zero on each finite cyclic subgroup $\langle g_1 \rangle$ and $\langle g_2 \rangle$.*

The cover of $B^\circ = \mathbb{P}^1(\mathbb{C}) \setminus \{0, 1, \infty\}$ with fibre $\mathcal{F}_D$ and these loop actions has a section exactly when $D = 1$.

Proof. The cocycle identity is block multiplication. If a cocycle is changed by a coboundary $B_g v - v$, its third coordinate at $g_\infty = (g_1 g_2)^{-1}$ does not change: the third row of $B_\infty - I$ is zero. Thus evaluation of that coordinate gives a homomorphism from H^1 to $\mathbb{Z}$, or to $\mathbb{Z}/D$. Its value on $[b]$ is -1 . Over $\mathbb{Z}$ the map $n \mapsto n[b]$ is therefore a split injection. Modulo D a multiple $k[b_D]$ can vanish only if $D \mid k$, while $D[b_D] = 0$ coefficientwise. The local fixed vectors $(2, -4, 0)$ and $(3, -3, 0)$ solve $b_j + B_j v = v$ for $j = 1, 2$, proving the two local trivializations.

A section of a finite covering over the connected base is a common fixed sheet of all monodromies: a chosen value continues uniquely along paths, and is single-valued precisely under that condition. The cusp changes z by -1 , so no fixed sheet exists for $D > 1$; for $D = 1$ the sole sheet is fixed. Equivalently an affine fixed point trivializes the cocycle. $\square$

This exhibits a locally split but globally nontrivial three-coordinate system, without replacing its arithmetic modulus by a continuous parameter. The source finite-filling maps $x \mapsto A_j x + v_j/m_j$ on a real torus are *different maps*: (2) is the linear monodromy restricted to $\gamma = 1$ modulo D , not a substitution for those filling translations.

2 All orbits and their compactified genera

Lemma 2.1 (Two shears and their central commutator). *The source words $H = [A_2^2, A_1]$ and $V = A_2 H A_2^{-1}$ induce*

$$H(x, y, z) = (x, y + 6, z + x - 6), \quad V(x, y, z) = (x - 6, y, z + y + 6).$$

Their commutator is

$$H V H^{-1} V^{-1}(x, y, z) = (x, y, z - 12). \quad (3)$$

These are identities over $\mathbb{Z}$, and hence in every residue ring.

Proof. Multiplication in (1) gives

$$H = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & 1 & 0 & 1 \end{pmatrix}, \quad V = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -6 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 6 & 0 & 1 & 1 \end{pmatrix}.$$

Their inverse affine formulas are $(x, y - 6, z - x + 6)$ and $(x + 6, y, z - y - 6)$. Substitution proves (3). $\square$

Theorem 2.2 (Complete odd-level classification). *If $3 \nmid D$, the action on $\mathcal{F}_D$ is transitive. If $3 \mid D$, it has exactly two orbits*

$$O_0 = \{x = y = 0 \pmod{3}\}, \quad O_1 = \mathcal{F}_D \setminus O_0,$$

of sizes $D^3/9$ and $8D^3/9$. The associated compact connected branched covers of $\mathbb{P}^1$ have genera

$$g(D) = \frac{5D^3 - 12D^2 - 17D + 24}{24}, \quad (3 \nmid D), \quad (4)$$

$$g_0(D) = \frac{5D^3 - 12D^2 - 81D + 216}{216}, \quad g_1(D) = \frac{5D^3 - 12D^2 - 9D + 27}{27} \quad (3 \mid D). \quad (5)$$

Every cusp cycle has length exactly D .

Proof. If $3 \nmid D$, both 6 and 12 are units modulo the odd integer D . The powers of H, V reach arbitrary x, y , and their central commutator then adjusts z arbitrarily. This proves transitivity with no assumption that D is prime.

Suppose $3 \mid D$. Modulo three the planar actions are $(x, y) \mapsto (y, -x - y)$ and $(x, y) \mapsto (-y, x)$. They fix zero and are transitive on the eight nonzero pairs. For an explicit verification, the words

$$1, a_1, a_2, a_2 a_1, a_2^2, a_2^2 a_1, a_2^3, a_2^3 a_1$$

send $(0, 1)$ respectively to $(0, 1), (1, 2), (2, 0), (1, 1), (0, 2), (2, 1), (1, 0), (2, 2)$. The powers of H, V then realize every planar change divisible by three, since $6(\mathbb{Z}/D) = 3(\mathbb{Z}/D)$. It remains to adjust z while restoring x, y . When $x = 0 \pmod{3}$, choose n with $6n = -x \pmod{D}$. The word $H^n a_\infty$ fixes x, y and shifts z by $c = -1 + n(x - 6)$, which is $-1 \pmod{3}$. Together with the central shift -12 , this generates every z -translation, because $\gcd(c, 12, D) = 1$. For a nonzero planar residue first use the listed words to reach $(0, 1)$ modulo three, apply this adjustment, and conjugate back. The two indicated orbits are thus complete.

A cusp iterate is $(x, y + nx, z - n)$, so its length is D . Solving the fixed-point equations gives

$$\text{Fix}(a_1) = \{(2, -4, z)\}, \quad \text{Fix}(a_2) = \text{Fix}(a_2^2) = \{(3, -3, z)\}.$$

The last equality follows from $a_2^2(x, y) = (-x + 6, -y - 6)$ and invertibility of 2. Thus a_2 has no cycles of length two. If $3 \nmid D$, the three cycle counts are

$$c_1 = (D^3 + 2D)/3, \quad c_2 = (D^3 + 3D)/4, \quad c_\infty = D^2.$$

If $3 \mid D$, all a_1 fixed points lie in O_1 and all a_2 fixed points in O_0 . For an orbit of size n with f_1, f_2 fixed points,

$$c_1 = f_1 + (n - f_1)/3, \quad c_2 = f_2 + (n - f_2)/4, \quad c_\infty = n/D.$$

Use $(n, f_1, f_2) = (D^3/9, 0, D)$ and $(8D^3/9, D, 0)$. The covering exists by the permutation-triple/Riemann-existence correspondence [Sijlsing and Voight(2014), Introduction and §1]; concretely, copies of a slit sphere are glued according to the permutations, and a length- e cycle is filled by the local map $w \mapsto w^e$. This also specifies the complex structure. The bicoloured cell structure has $c_1 + c_2$ vertices, n edges, and c_∞ faces, so $2 - 2g = c_1 + c_2 + c_\infty - n$. Substitution proves (4)–(5). $\square$

The finite permutations and their cyclic order are a constructive specification of the covers. The certificate does not purport to give rational equations for every map, or to prove that every component is defined over $\mathbb{Q}$ rather than $\overline{\mathbb{Q}}$. For $D = 3$ the two components have degrees $(3, 24)$ and genera $(0, 1)$; for $D = 5$ there is one component of degree 125 and genus 11. These numbers are not identifications with the Albert algebra or Leech lattice.

Proposition 2.3 (Functorial modulus maps). *For $D \mid E$, reduction $\mathcal{F}_E \rightarrow \mathcal{F}_D$ is equivariant and surjective, with fibres of size $(E/D)^3$. For coprime odd D, E ,*

$$\mathcal{F}_{DE} \xrightarrow{\sim} \mathcal{F}_D \times \mathcal{F}_E$$

is equivariant in all three coordinates and all three monodromies. An inverse coordinate is $a + D((b - a)D^{-1} \bmod E) \bmod DE$. The cocycles, local systems and completed covers have the corresponding maps. Products are taken over the same base and the same candidate label.

Proof. All maps are integral polynomials, so commute with reduction. The coordinate CRT formula proves both inverse compositions and the fibre counts. Extension across a puncture follows in local coordinates: a cycle of length E maps to one of length D by $w \mapsto w^{E/D}$, with its marked branch included. No CRT assertion is made between unrelated arithmetic leaves. $\square$

3 The full labelled arithmetic atlas

Fix a prime $p = 4m + 1$. The complete first-denominator interval is

$$m + 1 \leq a \leq 3m, \quad R = 4a - p.$$

It has $3 \leq R \leq 2p - 3$ and $\gcd(a, R) = \gcd(p, R) = 1$. For $a = \prod_q q^{e_q}$ retain every vector $f \in \prod_q \{0, \dots, 2e_q\}$ and $u = \prod_q q^{f_q}$. The inverse is $f_q = v_q(u)$. The channel formulas are

$$Q_E = \left(a, \frac{pa + a^2/u}{R}, \frac{pa + p^2u}{R} \right), \quad (6)$$

$$Q_M = \left(a, \frac{p(a + a^2/u)}{R}, \frac{p(a + u)}{R} \right). \quad (7)$$

For all raw orders, use the disjoint labels $(E, u, 0), (E, u, 1), (M, u, 0)$; the exterior bit reverses the last two positions. The middle reversal is already $u \mapsto a^2/u$, so another middle bit would duplicate it. The first coordinate need not be the smallest at an arbitrary rational candidate; it was the smallest only in the completeness argument below.

Lemma 3.1 (Exact rational and integral return). *Every such candidate is a positive rational decomposition of $4/p$. Its last two entries have the same reduced denominator*

$$D_E = \frac{R}{\gcd(R, 4u + 1)}, \quad D_M = \frac{R}{\gcd(R, u + a)}.$$

In particular D is odd and prime to p , and $D = 1$ exactly at an integral candidate. ES holds at p exactly when some full-atlas candidate has $D = 1$. The number of full ordered labels is at most $\frac{3}{2}(p-1)^2$.

Proof. Writing $Q = (a, Y, Z)$, substitution gives $(RY - pa)(RZ - pa) = p^2a^2$; both factors are positive. Expansion proves the rational reciprocal identity. For E , multiplying the Y numerator by the unit u gives $a^2(4u+1)$ modulo R , and the Z numerator is $4a^2(4u+1)$ modulo R . For M , the analogous numerators reduce, up to units, to $u+a$. Their gcds with R therefore agree exactly, proving the minimal denominators.

For any positive integral solution, move a smallest denominator to the first position and retain the permutation. Then $p/4 < a \leq 3p/4$, so it is in this interval. The positive factors $d_Y = RY - pa, d_Z = RZ - pa$ multiply to p^2a^2 . As $p \nmid a$, $v_p(d_Y)$ is 0,1 or 2. The three inverse labels are respectively

$$(E, a^2/d_Y, 0), \quad (M, pa^2/d_Y, 0), \quad (E, d_Y/p^2, 1).$$

The congruence $d_Y \equiv -pa \pmod{R}$ gives the corresponding gate by multiplication by units. Conversely an integral candidate is a solution. Finally $\tau(a^2) \leq 2a-1$ by pairing divisors around a ; hence $3 \sum_{a=m+1}^{3m} \tau(a^2) \leq 24m^2 = \frac{3}{2}(p-1)^2$. This is the full-shell form of the inherited Type I/II divisor parametrizations [Elsholtz and Tao(2013), §2]; it is not claimed as a new parametrization. $\square$

For every rational candidate keep

$$g = \gcd(a, u), \quad h = g^2/u, \quad r = u/g, \quad s = a/g,$$

so h, r, s are positive integers, $a = hrs$, $u = hr^2$ and $\gcd(r, s) = 1$. The quotients $\kappa = (pr+s)/R$ and $\lambda = (r+s)/R$ remain rational until the gate passes. In the unswapped order the inverse is $u = a^2/(RY - pa)$ for E and $u = pa^2/(RY - pa)$ for M . Clearing $X = DQ$ gives a positive integral solution at pD , not at p :

$$F_p(X) = 4X_0X_1X_2 - p \sum_{i < j} X_iX_j = p(D-1) \sum_{i < j} X_iX_j.$$

The prime, the scale and the raw order therefore remain marks.

4 Arithmetic obstruction groups become geometric cocycles

For a candidate α let $g_E = 4u+1$ or $g_M = u+a$, $h_0 = \gcd(R, g_c)$, and $D = R/h_0$. Its cyclic obstruction group is the *actual* subgroup

$$O_\alpha = \langle [g_c]_R \rangle \leq \mathbb{Z}/R.$$

The map $\mathbb{Z}/D \rightarrow O_\alpha$, $k \mapsto kg_c \pmod{R}$, is a bijection. On a representative $v \in O_\alpha \cap \{0, \dots, R-1\}$ its inverse is

$$k = \frac{v}{h_0} (g_c/h_0)^{-1} \pmod{D};$$

modulo one the inverse is the unique zero class. Each prime power is retained by

$$v_\ell(D) = \max\{v_\ell(R) - v_\ell(g_c), 0\}.$$

Combining this inverse with Theorem 1.1 gives the *faithful* map

$$O_\alpha \xrightarrow{\sim} \langle [b_D] \rangle, \quad kg_c \longmapsto k[b_D]. \quad (8)$$

The inverse first reads minus the third cusp evaluation modulo D , then multiplies by g_c modulo R . Thus the denominator obstruction is not merely assigned a similarly sized group: it is explicitly identified with a split cyclic subgroup of the geometric cocycle group.

Definition 4.1. The *full denominator-labelled cover* at p is the disjoint union, over every ordered atlas label α , of the covers with fibres $\{\alpha\} \times \mathcal{F}_{D_\alpha}$ and actions $(\alpha, v) \mapsto (\alpha, a_j v)$. Its decoration includes the prime, shell, factorization, exponent vector, channel, swap, h, r, s , channel quotient, rational triple, clearing scale and all local losses.

Theorem 4.2 (Complete failure object). *ES fails at p if and only if this full labelled cover has no section. Equivalently, every summand has a nonzero cocycle class of exact order $D_\alpha \geq 3$, and every cusp permutation has no fixed sheet. The compactified components, degrees, ramification and genera of the complete object are given by Theorem 2.2, applied with their full arithmetic multiplicities.*

Proof. Over the connected base a section has a constant discrete candidate label. Theorem 1.1 says its value in that summand is single-valued exactly when $D_\alpha = 1$. Lemma 3.1 proves the arithmetic biconditional. The finite connected components are precisely the orbits, so Theorem 2.2 supplies the stated complete classification. $\square$

The cover degree is bounded by $\frac{3}{2}(p-1)^2(2p-3)^3$. The small input consisting of all labelled D_α and the three formulas specifies every sheet and every edge; it is not necessary to expand that many sheets to validate the atlas. An actual global negative certificate would have to provide *all* labels and a nonzero prime-power loss on each. No global negative example is supplied. The algorithm returns an existing positive witness on all four primes used in its full-atlas tests.

The previous operator $H_p = \text{diag}((D_\alpha - 1)/2)$ has a separate meaning. Its inverse, when all $D > 1$, multiplies the α -th coordinate by $2/(D_\alpha - 1)$. Every geometric loop permutation is invertible regardless of whether $D = 1$. Neither assertion is the inverse of the other. The same denominator connects them, with its label retained.

Restricted prototype. At $p = 241, a = 64, R = 15$ every divisor is $u = 2^e, 0 \leq e \leq 12$. For each canonical channel, complete modular arithmetic gives $D = 3$ four times, $D = 5$ six times, and $D = 15$ three times. Including the exterior swap gives 39 labels, with multiplicities 12,18,9 respectively. The associated cover has

$$32949 \text{ sheets,} \quad 60 \text{ connected components,} \quad \sum g = 5448.$$

All 39 summands are sectionless. This is an actual complete negative *shell*, not a false prime: other shells at 241 succeed. Taking the gcd of its denominators gives one and would destroy the leafwise obstruction. The validator rejects a restricted certificate, a deleted row, or a full atlas containing a successful candidate.

Existing arithmetic transports lift, but do not erase their labels. Exterior order reversal and middle complementation preserve D . The retained relation from $E(u)$ to $M(u')$, where $u, u' \mid a^2$ and $u' \equiv pu \pmod{R}$, preserves D because $u' + a \equiv a(4u + 1) \pmod{R}$ and a is a unit. Thus every such labelled edge has the explicit equivariant cover map $(\alpha, v) \mapsto (\alpha', v)$, with reverse edge as inverse. The endpoint labels and the whole chosen path are retained; forgetting them would make this cover-valued functor nonfaithful on paths. The 39-state restricted graph above has three components, of sizes 12,18,9 at levels 3,5,15. The certificate lists all edges. These particular transports are defect-preserving, not a claimed strict descent. Cross-shell arithmetic still needs an additional argument; arbitrary independent choices of D_α are not substituted for the simultaneously calculated values from one prime.

5 What the inspected S^6 construction actually contains

The live workbench revision read for this tranche is 8cfbd18f3e398c533e1acabfba98b4a29975000a. Its S_6 guide distinguishes the original construction from later scoped reconstructions

[S6 workbench(2026)]. We read the five-page key-advances TeX and the local-source passages identified below, not the entire externally hosted 1,079-file, 174-MB frozen project. The source manuscript was read through its public PDF; attempts to download it into this runtime failed DNS resolution. No full-file hash or independent global verification is invented.

The manuscript circulated by Levent Alpöge describes a complex threefold $X \rightarrow \mathbb{P}^1$, with complex two-torus fibres off $0, 1, \infty$. Engel's separate exposition attributes the manuscript to Claude, communicated by Alpöge, and offers geometric explanation rather than a certification asserted by this tranche [Source(2026), Engel(2026)]. The following are the precise source objects, not conclusions about an ES counterexample.

5.1 Periods and actual duality

The source integral matrices on $V = \mathbb{L}^*$ are

$$T_1 = \begin{pmatrix} 1 & 0 & -6 & 2 \\ 0 & -1 & 1 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad T_2 = \begin{pmatrix} 1 & 6 & 0 & -3 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

The inverse-transpose relation $A_j = (T_j^{-1})^t$ proves $\langle T_j v, A_j \ell \rangle = \langle v, \ell \rangle$ exactly. The ordered period map is

$$\Pi = \begin{pmatrix} 6\mu & \tau & 1 & 0 \\ \beta & \mu & 0 & 1 \end{pmatrix} : \mathbb{R}^4 \rightarrow \mathbb{C}^2, \quad \Im \tau > 0, \quad \Im \beta - \frac{6(\Im \mu)^2}{\Im \tau} < 0.$$

These conditions make it real-linearly invertible. On the covering upper half-plane its equivariance is $\Pi(gz) = R_g(z)\Pi(z)\rho_V(g)^t$, with $R_g(z) \in \mathrm{GL}_2(\mathbb{C})$. The period functions obey, among others,

$$\begin{aligned} \tau g_1 &= (\tau - 1)/\tau, & \tau g_2 &= -1/\tau, & \mu g_1 &= (1 - \mu)/\tau, & \mu g_2 &= 1 + \mu/\tau, \\ \beta g_1 &= \beta + 2 - 6(1 - \mu)^2/\tau, & \beta g_2 &= \beta - 3 - 6\mu^2/\tau. \end{aligned}$$

At the two finite centres $\tau = e^{\pi i/3}$ and i ; at the cusp the specified μ and $\beta + \tau$ are bounded. The common invariant covector is the γ used in the affine slice.

On a fixed regular fibre the map

$$(\mathbb{L}/D\mathbb{L}) \longrightarrow (\mathbb{C}^2/\Pi\mathbb{L})[D], \quad [v] \longmapsto [\Pi v/D] \quad (9)$$

is a bijection. If $D\zeta = \Pi v$ in the torus, invertibility of Π recovers v modulo D ; this proves both directions. Equivariance of the periods makes (9) commute with A_j . Its restriction to $\gamma(v) = 1$ is exactly the cover in this tranche. Every such point has exact torsion order D . This is a literal comparison to the regular period local system.

5.2 Opposite-side gluing and the finite fillings

The cusp fibre has normalization dP_6 , the blow-up of $\mathbb{P}^2$ at three noncollinear points. Its six boundary curves form a hexagon. Three pairs of opposite curves are identified. The resulting fibre W has three double curves and two triple points, with local total-space models $t_c = z_1$, $z_1 z_2$, or $z_1 z_2 z_3$. The normalization has respectively one, two, or three points over those strata. The side-pairings are reversible on their marked branches; the normalization-to-quotient map is not injective there. Retaining a branch label is necessary to lift a point back.

At the finite centres the source uses ramified base coordinates $t_j = s_j^{m_j}$, $(m_1, m_2) = (3, 4)$, and free affine quotient actions $\xi \mapsto A_j \xi + v_j/m_j$ on the real period torus, coupled with the base rotation. The complete deck-group relations are

$$\Gamma_j = \langle \mathbb{L}, G_j \mid G_j t_\ell G_j^{-1} = t_{A_j \ell}, \quad G_j^{m_j} = t_{v_j} \rangle.$$

The source-side attachment kernel is the central infinite cyclic group $\langle \tilde{G}_j^{m_j} t_{-v_j} \rangle$. The finite central product covers have degrees nine and eight, not three and four. Their kernels and affine translations are retained in the workbench key-advances §3.

5.3 The conjugate term in a period comparison

Another literal source “mirror” is a real-linear comparison of nearby periods, not a holomorphic involution. Put $T = \Im \tau > 0$, $m = \Im \mu$, $d = \Im \beta_{c_*} - 6m^2/T < 0$, and $P_* = \begin{pmatrix} 0 & 0 \\ -m/T & 1 \end{pmatrix}$. For $\beta_c = \beta_{c_*} + \delta$ the workbench gives

$$\zeta \mapsto \left(I + \frac{\delta}{2id} P_* \right) \zeta - \frac{\delta}{2id} P_* \bar{\zeta}, \quad \det_{\mathbb{R}} = \frac{d + \Im \delta}{d} > 0.$$

Both the inverse-transpose duality, the opposite-side identifications, and this conjugate term are exact operations. They are not interchangeable. The workbench further records a nonzero Kodaira–Spencer deformation class; we do not reprove that global statement here.

6 The completed filling presentation and the denominator

The source computes a cyclic fundamental group with order

$$n = |12\ell_0 - 4\ell_1 - 3\ell_2|,$$

for admissible translations, and claims that the original choice $(\ell_0, \ell_1, \ell_2) = (0, 1, -1)$ has the homology and smooth type of S^6 . The admissibility conditions are explicit: $3 \nmid \ell_1$ and ℓ_2 odd [Source(2026), Proposition 5.6 and §7]. Their geometric justification belongs to that source. The following presentation calculation is independent integer algebra.

Proposition 6.1 (Exact cyclic presentation comparison). *For each odd D prime to three, choose*

$$\ell_0 = 0, \quad \ell_2 = \begin{cases} 1 & D \equiv 1 \pmod{4}, \\ -1 & D \equiv 3 \pmod{4}, \end{cases} \quad \ell_1 = -\frac{D + 3\ell_2}{4}.$$

These choices are admissible and have $12\ell_0 - 4\ell_1 - 3\ell_2 = D$. The relation matrix

$$\begin{pmatrix} -\ell_1 & 3 & 0 \\ -\ell_2 & 0 & 4 \\ -\ell_0 & 1 & 1 \end{pmatrix}$$

has Smith invariants $(1, 1, D)$. Conversely every admissible nonzero value of $12\ell_0 - 4\ell_1 - 3\ell_2$ is prime to six.

Proof. The numerator defining ℓ_1 is divisible by four. Modulo three it is D , so $3 \nmid \ell_1$. The other assertions about the parameters follow by substitution. The determinant has absolute value D ; two minors from the last row and the first two rows give 3 and -4 , so the gcd of all two-by-two minors is one, yielding the Smith invariants. Mapping the three generators to $1, \ell_1/3, \ell_2/4$ modulo D annihilates the relations and is onto. The inverse sends $1 \bmod D$ to the first generator. An admissible value is odd modulo two and nonzero modulo three, proving the converse. The source’s original signed value is -1 , not $+1$. $\square$

Thus a denominator cyclic group can be compared to the completed-source presentation for $\gcd(D, 6) = 1$. Denominators divisible by three cannot be assigned such a free filling merely by choosing other integers in that same $(3, 4)$ family. The finite torsion-cover construction still works for them and has the two orbits already calculated. Neither comparison is an isomorphism between a finite ES atlas and a compact six-manifold.

7 A definite obstruction to one putative isomorphism

Proposition 7.1 (Genus survives the natural base change). *Take a positive-genus compact component $C_D \rightarrow \mathbb{P}^1$ of the cover, and a connected resolution of a dominant component of the normalized fibre product $X \times_{\mathbb{P}^1} C_D$, when the source compact fibration is used. Its fundamental group surjects onto $\pi_1(C_D)$, so its first Betti number is at least $2g(C_D)$. It is not diffeomorphic to S^6 .*

Proof. A loop of the compact base can be moved off the finitely many special points. Over that complement the pullback is a proper submersion with connected torus fibres. Lift the loop from a chosen point; connect its endpoint back to its start within the fibre. The resulting closed loop maps to the original loop, proving the surjection. The same open submersion is unaffected by normalization or a resolution that is an isomorphism over it. Abelianization gives the Betti inequality. $\square$

For every odd $D > 1$ at least one component has positive genus, by the formulas already proved. This only excludes this particular pullback as an S^6 copy. It does not exclude another carefully specified map, and it is not a contradiction to ES falsity: ordinary successful primes also have many rejected leaves with such components.

A different naive comparison already lies on a boundary. The native coefficient form of the regular period model is

$$G(t, m, b) = \begin{pmatrix} 2t & 2m \\ 2m & b/3 \end{pmatrix}, \quad \det G = \frac{2}{3}(tb - 6m^2).$$

Substituting the exact divisor relation $uv = a^2$ by $(t, m, b) = (u/2, a/2, 3v)$ gives $G = \begin{pmatrix} u & a \\ a & v \end{pmatrix}$ with determinant zero. Regular periods require the strict negative determinant, so this substitution is not a regular-fibre embedding. Our torsion comparison uses the valid period lattice and the integer level D , not that singular substitution.

8 The level-three group, with every matrix retained

At $D = 3$ the generated matrix group has the form

$$\left\{ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & a & b & 0 \\ 0 & c & d & 0 \\ \chi(A) & r & s & 1 \end{pmatrix} : A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{F}_3), \quad r, s \in \mathbb{F}_3 \right\}.$$

Here $\chi : \mathrm{SL}_2(\mathbb{F}_3) \rightarrow \mathbb{F}_3$ is the character taking the planar part of A_1 to 1 and that of A_2 to 0. For a finite exact proof, $Q = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and its conjugate by $P = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$ generate the eight-element quaternion subgroup; P normalizes it and has order three. Hence $\mathrm{SL}_2(\mathbb{F}_3) = Q_8 \rtimes \langle P \rangle$, and the exponent of P defines χ . The kernel translations with bottom rows $(0, 1, 0, 1)$ and $(0, 0, 1, 1)$ are respectively the source words

$$A_2 A_2 A_1 A_2 A_2 A_1 A_1, \quad A_1 A_1 A_2 A_2 A_1 A_2 A_2 \pmod{3}.$$

Multiplication verifies them. They generate all nine row translations; together with the projected 24 matrices they give exactly the displayed 216 matrices. Removing (r, s) gives an explicit splitting. The group is therefore $(\mathbb{F}_3^2)^* \rtimes \mathrm{SL}_2(\mathbb{F}_3)$, isomorphic to $\mathrm{ASL}_2(\mathbb{F}_3)$ using the invariant alternating pairing to identify the dual module. This identification is verified as a matrix-group statement, not inferred from the order 216 alone. No identification of its 27-point action with the preceding Albert coordinates is asserted.

9 Reproducibility and precise scope

The standalone verifier recomputes primality, factorizations, every full exponent box, every ordered rational return, all local denominator losses, and the actual finite affine permutations at the stated levels. It completely enumerates levels 1, 3, 5, 7, 9, 15, 21, 25, 27, 35, 45 and full arithmetic atlases at primes 5, 37, 241, 1201. Large covers attached to the arithmetic atlases are specified by exact formulas and proved passports; their D^3 sheets are not all expanded. The restricted negative example $p = 241, a = 64$ is checked separately and cannot pass the full-negative validator.

The code uses only the Python standard library and raises explicit errors, including in optimized execution. The certificate stores every source matrix and affine branch; the $D \leq 9$ permutations are supplied in full. The source topology and analytic period-existence theorem were not independently rerun, nor was a Lean build performed. A precise dependency plan accompanies this tranche. No global ES resolution or independently certified global complex structure on S^6 is claimed.

The orbit classification, its exact genus formulas, and the marked cyclic-obstruction comparison were derived in this investigation. A content-level search located the source monodromies, the standard permutation-cover correspondence and classical finite group framework, but no exact antecedent for these combined formulas was established. This is not a historical-priority claim. The compact companion preprint isolates those calculations rather than republishing a long historical corpus. No repository was modified and no external deposit was made.

References

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